

Software Frameworks for Explainable AI in Time Series Classification: A Systematic Review

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Supplementary Material

Appendix A

Table A.1: XAI methods of frameworks. **Compatibility:** ✓ indicates method is specific for time series, otherwise marked as – (not explicitly stated); **Dimensionality:** supports uni- (U), multivariate (M) time series or both (B); **Frequency:** ✓ generates explanations for the frequency and time-frequency domain; **Frameworks:** ✓ indicates that the framework implements the method, (✓) if the framework wrapped or reused implementations of other frameworks, and ✗ otherwise. Abbr.: Counterfactuals (CF)

Methods	Compatibility	Dimensionality	Frequency	Quantus	Frameworks					
					TSInterpret	time_interpret	tsCaptum	XTSC-Bench	SIGN-XAI-2	
BetaSmooth	-	-	✗	✗	✗	✗	✗	✗	(✓)	
Conductance	-	-	✗	(✓)	✗	✗	✗	✗	✗	
Deconvolution	-	-	✗	(✓)	✗	✗	✗	✗	(✓)	
Deep Learning Important Features (DeepLIFT)	-	-	✗	(✓)	(✓)	✗	✗	(✓)	✗	
DeepLIFT Shapley Additive Explanations (SHAP)	-	-	✗	(✓)	(✓)	✗	✗	(✓)	✗	
DFT-LRP	✓	U	✓	✗	✗	✗	✗	✗	✓	
Discretized Integrated Gradients	-	-	✗	✗	✗	✓	✗	✗	✗	
Excitation Backpropagation	-	-	✗	✗	✗	✗	✗	✗	(✓)	
GeodesicIntegratedGradients	-	-	✗	✗	✗	✓	✗	✗	✗	
Gradient-weighted Class Activation Mapping (Grad-CAM)	-	-	✗	(✓)	(✓)	✗	✗	(✓)	✗	
Gradient × Input	-	-	✗	(✓)	✗	✗	✗	✗	✗	
GradientSHAP	-	-	✗	(✓)	(✓)	✗	✗	(✓)	✗	
Guided Backpropagation	-	-	✗	(✓)	✗	✗	✗	✗	(✓)	
Guided Grad-CAM	-	-	✗	(✓)	✗	✗	✗	✗	✗	
Integrated Gradients	-	-	✗	(✓)	(✓)	✗	✗	(✓)	(✓)	
InternalInfluence	-	-	✗	(✓)	✗	✗	✗	✗	✗	
Layer Activation	-	-	✗	(✓)	✗	✗	✗	✗	✗	
Layer-wise Relevance Propagation (LRP)	-	-	✗	(✓)	✗	✗	✗	✗	(✓)	
Saliency (Vanilla Gradients)	-	-	✗	(✓)	(✓)	✗	✗	(✓)	(✓)	
Sequential Integrated Gradients	-	-	✗	✗	✗	✓	✗	✗	✗	
SIGN	-	-	✗	✗	✗	✗	✗	✗	✓	
SmoothGrad	-	-	✗	(✓)	(✓)	✗	✗	(✓)	(✓)	
Testing with Concept Activation Vectors (TCAV)	-	-	✗	(✓)	✗	✗	✗	✗	✗	

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Table A.1 – *continued from previous page*

Methods	Compatibility	Dimensionality	Frequency	Frameworks					
				Quantus	TSInterpret	time_interpret	tsCaptum	XTSC-Bench	SIGN-XAI-2
Temporal Integrated Gradients	✓	B	×	×	×	✓	×	×	×
Tracing Gradient Descent with Checkpoints (TracInCP)	-	-	×	(✓)	×	×	×	×	×
Augmented Occlusion	✓	B	×	×	×	✓	×	×	×
BayesKernelSHAP	-	-	×	×	×	✓	×	×	×
BayesLIME	-	-	×	×	×	✓	×	×	×
DynaMask	✓	B	×	×	×	✓	×	×	×
Extremal Mask	✓	B	×	×	×	✓	×	×	×
Feature Ablation	-	-	×	(✓)	(✓)	✓	(✓)	(✓)	×
Feature Permutation	-	-	×	(✓)	×	×	(✓)	×	×
KernelSHAP	-	-	×	(✓)	×	×	(✓)	×	×
LEFTIST	✓	U	×	×	✓	×	×	(✓)	×
Local Interpretable Model-agnostic Explanations (LIME)	-	-	×	(✓)	×	×	(✓)	×	×
Local Outlier Factor (LOF)-LIME	-	-	×	×	×	✓	×	×	×
LOF-KernelSHAP	-	-	×	×	×	✓	×	×	×
Occlusion	-	-	×	(✓)	(✓)	✓	×	(✓)	(✓)
Shapley Value Sampling	-	-	×	(✓)	(✓)	×	(✓)	(✓)	×
Temporal Augmented Occlusion	✓	B	×	×	×	✓	×	×	×
Temporal Occlusion	✓	B	×	×	×	✓	×	×	×
COMTE	✓	M	×	×	✓	×	×	(✓)	×
NUN-CF	✓	U	×	×	✓	×	×	(✓)	×
Shapelet explainer for time series (SETS)	✓	B	×	×	✓	×	×	(✓)	×
TSEvo	✓	B	×	×	✓	×	×	(✓)	×
Activations Visualization	-	-	×	×	×	×	×	×	×
Feature Importance in Time (FIT)	✓	B	×	×	×	✓	×	×	×
Layer/Neuron Activations	-	-	×	×	×	×	×	×	×
Reverse Time Attention model (RETAIN)	✓	B	×	×	×	✓	×	×	×
Time Forward Tunnel	✓	B	×	×	×	✓	×	×	×
Temporal Saliency Rescaling	✓	M	×	×	✓	×	×	(✓)	×

Table A.2: Evaluation metrics of frameworks. Legend: **Compatibility:** ✓ indicates metric is specific for time series, C if the metric is compatible with time series, and ✗ otherwise; **Quantus, time_interpret, XTSC-Bench:** ✓ indicates that the framework implements the metrics, (✓) if the framework wrapped or reused implementations of other frameworks, and ✗ otherwise.

Metrics		Compatibility	Quantus	time_interpret	XTSC-Bench
Perturbation-based	Accuracy	C	✗	✓	✗
	Avg-Sensitivity	C	✓	✗	(✓)
	Comprehensiveness	C	✗	✓	✗
	Continuity	✗	✓	✗	✗
	Consistency	C	✓	✗	✗
	Cross_entropy	C	✗	✓	✗
	Faithfulness Correlation	C	✓	✗	(✓)
	Faithfulness Estimate	C	✓	✗	(✓)
	Infidelity	✗	✓	✗	✗
	Iterative Removal of Features (IROF)	✗	✓	✗	✗
	Lipschitz_max	C	✗	✓	✗
	Local Lipschitz Estimate	C	✓	✗	✗
	Log_odds	C	✗	✓	✗
	Max-Sensitivity	C	✓	✗	(✓)
	Mean absolute error (MAE)	C	✗	✓	✗
	Mean squared error (MSE)	C	✗	✓	✗
	Monotonicity (Arya)	C	✓	✗	(✓)
	Monotonicity (Nguyen)	C	✓	✗	(✓)
	Pixel Flipping	C	✓	✗	✗
	Region Perturbation	C	✓	✗	✗
	Relative Input Stability	C	✓	✗	✗
	Relative Output Stability	C	✓	✗	✗
	Relative Representation Stability	C	✓	✗	✗
	Remove and Debias (ROAD)	✗	✓	✗	✗
	Selectivity	✗	✓	✗	✗
	SensitivityN	C	✓	✗	✗
	Sufficiency ¹ by DeYoung et al. [2]	C	✗	✓	✗

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Table A.2 – *continued from previous page*

Metrics		Compatibility	Quantus	time_interpret	XTSC-Bench
Ground-Truth-based	Attribution Localisation	C	✓	x	x
	Area under precision curve (AUP)	C	x	✓	x
	Area under precision-recall curve (AUPRC)	C	x	✓	x
	Area under recall curve (AUR)	C	x	✓	x
	Area under the Receiver Operating Characteristic Curve (ROC-AUC)	C	✓	✓	(✓)
	Focus	x	✓	x	x
	MAE	C	x	✓	x
	Mask entropy	✓	x	✓	x
	Mask information	✓	x	✓	x
	MSE	C	x	✓	x
	Pointing Game	C	✓	x	(✓)
	Relevance Mass Accuracy	C	✓	x	(✓)
	Relevance Rank Accuracy	C	✓	x	(✓)
	Root mean squared error (RMSE)	C	x	✓	x
	Top-K Intersection	C	✓	x	x
Other	Complexity	C	✓	x	(✓)
	Effective Complexity	C	✓	x	x
	Efficient MPRT	C	✓	x	x
	Input Invariance	C	✓	x	x
	Model Parameter Randomization Test (MPRT)	C	✓	x	x
	Non-Sensitivity	C	✓	x	x
	Random Logit Test	C	✓	x	x
	Smooth MPRT	C	✓	x	x
	Sparseness	C	✓	x	x
	Sufficiency ¹ by Dasgupta et al. [1]	C	✓	x	x

¹ Two metrics share the name *Sufficiency* and are differentiated by the reference to their original publications.

References

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2. DeYoung, J., Jain, S., Rajani, N.F., Lehman, E., Xiong, C., Socher, R., Wallace, B.C.: ERASER: A benchmark to evaluate rationalized NLP models. In: Jurafsky, D., Chai, J., Schluter, N., Tetreault, J.R. (eds.) Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 4443–4458. Association for Computational Linguistics, online (2020). <https://doi.org/10.18653/V1/2020.ACL-MAIN.408>